

## Development of a Six Bar Mechanism Based Active Knee Orthosis for Post-Stroke Gait Rehabilitation

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**Abstract:** An active knee orthosis can support a patient's weight and assist the knee during stance phase for improved gait assistance. This paper presents the development of a lightweight active knee orthosis (AKO) for post-stroke gait rehabilitation that is capable of providing assistive torque in a pattern that is similar to the natural pattern of the human leg. The proposed system is designed to be lightweight and comfortable to use. The system utilizes a single series elastic actuator and is controlled using the patient-driven motion reinforcement (PDMR) control to provide assistive torque to the user's knee based on the force information gathered from its series elastic link. A pilot study was performed with one young healthy subject wearing an ankle weight (4 kg) on one leg, in order to identify the effects and potential implications of using the AKO for gait assistance of hemiplegic subjects. No discomfort was reported by the participant during the gait trials. Addition of the ankle weight reduced the knee movements, gait speed and gait balance, while increasing the knee range of motion asymmetry. While using the AKO, the subject was able to overcome much of the detrimental effects of the ankle weight and show improvements in majority of the outcome measures. Thus, the developed AKO exhibited promising outcomes in this pilot study that warrant further studies to better understand the implications of using this system during gait rehabilitation of stroke survivors.

**Keywords:** Exoskeleton; Gait Rehabilitation; Stroke Rehabilitation; Active Knee Orthosis.

### 1. INTRODUCTION

Gait rehabilitation is essential for recovery of gait ability and daily living competence [1]. An active knee orthosis (AKO) can support various knee functions for the chronic stroke survivors' rehabilitation, such as sit-to-stand exercise, over ground walking and stair climbing [2]. Especially, during the stance phase of gait, the AKO can support the patient's weight and assist the knee movement [2, 3]. Therefore, numerous such devices employing a variety of mechanisms have been developed, however their commercialization has been hindered by their high weight, high cost etc [4]. A representative commercial AKO is the Tibion Bionic Leg, which has a weight of 3.5 kg, but uses multiple actuators and a transmission for torque conversion [5], which makes the system complex and increases its cost.

S. Pyo et al [6] suggested an AKO based on a geared-five-link mechanism capable of performing different functions during the swing (torque assistance) and stance (locking, fall prevention) phases of gait. Building upon this design, we developed a novel system that combined an AKO and a haptic cane to provide assistance during overground walking [2]. The function of the AKO developed for that research was focused on swing phase of gait, i.e. it did not include the locking function. It employed a 6 bar mechanism actuated by one motor that was used in a series elastic actuation (SEA) configuration. The 6 bar mechanism emulates the human knee's torque

characteristics; i.e. generation of low amounts of torque when the knee angle is small and high amounts when the knee angle is large. The elastic component of the SEA incorporated in that system functioned to estimate the intention of the user. However, that system weighed 6.2 kg, which limited its usability for patients such as stroke survivors. In this paper, we present a new version of the previously developed AKO, which has been modified to reduce its weight and increase its comfort, in order to improve its usability. In addition, a pilot test was conducted with a healthy subject with simulated gait asymmetry to confirm that the developed device can improve the knee range of motion symmetry ratio by providing assistive knee torque during walking.

### 2. SYSTEM DEVELOPMENT

The primary design objective of the AKO is to provide assistive torque to help the user perform lower limb movements. Such assistance is required especially in the stance phase when the quadriceps muscle cannot provide the knee moments necessary to maintain the stance control. The developed AKO is a six-link system composed of two prismatic joints and four revolute joints (Fig. 1). The parameters of the links are given in Table 1. Also mentioned in the Table are  $F_s$  and  $v_s$ , which are the desired force and velocity of the actuator. These values are necessary to determine the required actuator power.

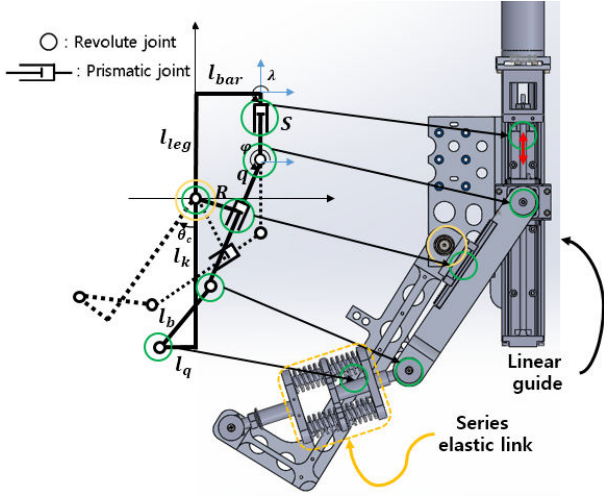


Fig. 1. Kinematic configuration of the AKO

Table 1. Link parameters of the AKO

|          | Parameter | Value    | Parameter | Value  |
|----------|-----------|----------|-----------|--------|
| Linkages | $l_k$     | 250 mm   | $q$       | 190 mm |
|          | $l_q$     | -80 mm   | $l_b$     | 165 mm |
|          | $l_{bar}$ | 70 mm    | $l_{leg}$ | 100 mm |
|          | $R$       | 35 mm    |           |        |
| Actuator | $v_s$     | 300 mm/s | $F_s$     | 500 N  |

Link kinematics of our AKO can be determined from the geared-five-link AKO developed by S. Pyo et al [6]. Thus,  $l_b$  of the modified AKO is given as:

$$l_b^2 = (l_{bar} + S \cos \lambda - l_q \cos \theta_c + l_q \cos \varphi + l_k \sin \theta_c)^2 + (l_{leg} + l_k \cos \theta_c + S \sin \lambda + l_q \sin \theta_c + q \sin \varphi)^2 \quad (1)$$

From (1), angular velocity equation of link  $l_k$  driven by a linear actuator can be derived as:

$$\dot{\theta} = \frac{\partial \theta_c}{\partial s} \cdot \frac{ds}{dt} \quad (2)$$

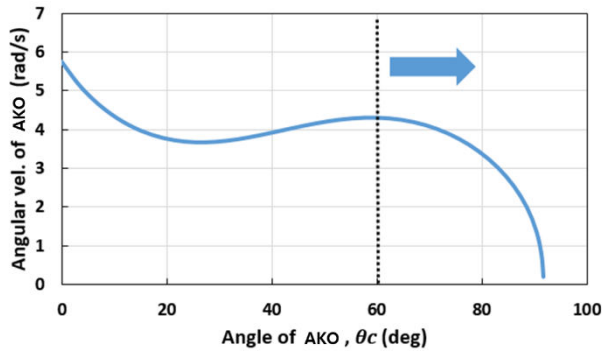


Fig. 2. Angular velocity of AKO vs. angle of AKO

Fig. 2 shows the angular velocity of the AKO vs. angle of the AKO knee joint  $\theta_c$ . Here, we can clearly observe the kinematic characteristics of our design, which exhibits reduced desired angular velocity as the knee angle goes beyond 60 degree. This behavior is similar to that of the human knee mechanism: i.e. low velocity and

high torque at high knee angle, like when standing up from a chair, and high velocity and low torque at low knee angle, like during walking. As shown in Fig. 3, user comfort during walking is increased through use of a prismatic joint between link  $l_k$  and the orthosis structure that attaches to the user's calf. This provides a limited amount of compliance for the user and allows them to walk comfortably even when there is a small difference between their and the AKO's knee angle trajectories.

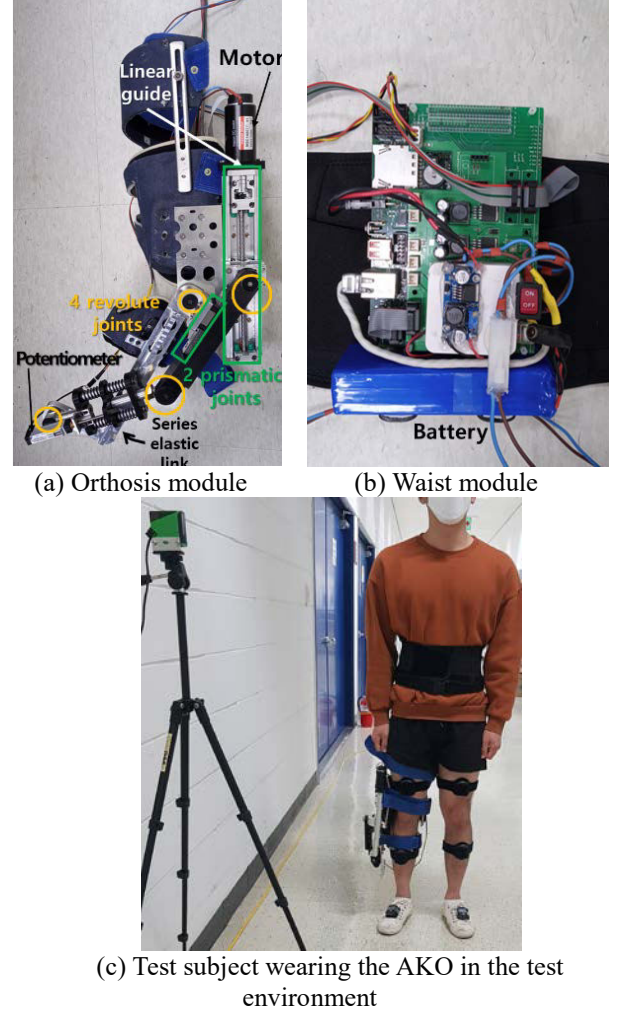


Fig. 3. The developed AKO and its testing

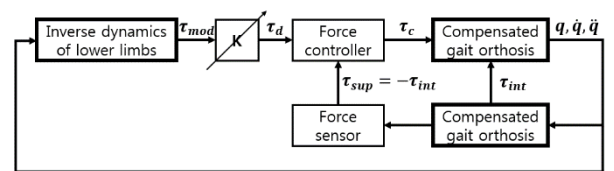


Fig. 4. Block diagram of the PDMR control logic [8].

Fig. 4 presents the block diagram of the control logic implemented in the AKO. In this system, in order to ensure a comfortable user experience, we have implemented the Patient-Driven Motion Reinforcement (PDMR) scheme, similar to that used in the Lokomat [7]. In this control logic, the torque is generated in accordance with the user's intention and  $\tau_{sup}$  is used to compensate

the force used by the wearer to assert their intention. The force sensor block shown in Fig. 4 that is used to sense the user's intention is realized in the form of the potentiometer sensor based Series Elastic Link shown in Fig. 3 (a). Fig. 5 presents the flow chart of the low-level controller used in the AKO, while the hardware specifications of the system are presented in Table 2.

The system's weight has been reduced through a number of measures. A single Maxon DC motor that does not require the use of a transmission actuates the entire system. Wherever possible, the system parts have been manufactured using molded or 3D printed plastic. Parts that could not be manufactured using plastic have been manufactured using lightweight alloys so that the desired strength can be achieved without increasing the weight by a large amount. In order to reduce the disruptive effect of having a large mass attached to a person's extremity, the system electronics (battery, motor driver, controller board and communication device) have been mounted on an elasticated belt with Velcro fasteners that the user can easily wear on their waist.

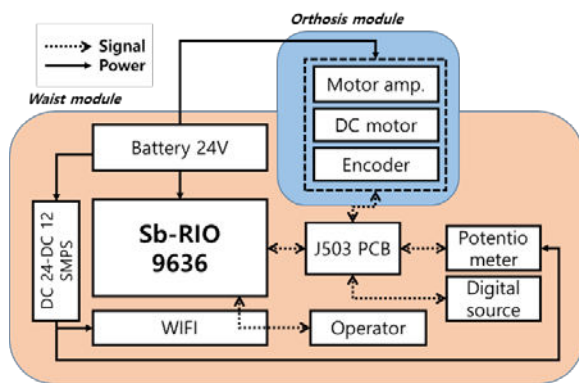


Fig. 5. Flow chart of the low-level controller

Table 2. Specifications of the AKO

|                         |                                                    |
|-------------------------|----------------------------------------------------|
| System weight           | 3.73 kg (Orthosis: 2.69 kg, Waist module: 1.04 kg) |
| DC Motor and Controller | Maxon DC motor 150W, ESCON 50/5                    |
| Micro controller        | Sb-RIO 9636 with LabVIEW                           |
| Linear actuator         | AKE20 (THK)                                        |
| Displacement sensor     | Potentiometer                                      |
| Encoder resolution      | 1024                                               |
| Max. speed              | 330 deg/s                                          |

### 3. PILOT TEST

We performed a pilot study with one young healthy subject in order to identify the effects and potential implications of using our developed AKO. Since the developed AKO is a rigid system that applies moments to the knee, it was deemed necessary to validate the system function and verify its effects through a pilot study before conducting clinical trials with a population suffering with gait disabilities. A 24 year old male

participated in the gait trials (height: 169 cm; weight: 66 kg). The subject had no history of musculoskeletal or neurological disorders.

#### 3.1 Protocol

The subject was introduced to the system and its use through verbal explanation. In order to emulate the weakness and gait deficiencies found in a hemiparetic patient, an ankle weight of 4 kg was attached to the subject's leg [8]. The subject wore the ankle weight and the AKO on the right leg. The AKO was wirelessly monitored and controlled by the operator using a PC based interface software. The trial conditions were as follows:

- Normal Walking with no ankle weight (NW)
- Walking with Ankle weight (WA)
- Walking with Ankle weight and AKO (WAA)

The subject walked a distance of 10m in each trial and performed two trials under each condition. The data recorded during the two iterations of each trial condition were averaged. 2-3 min seated rest was provided to the subject between trials.

#### 3.2 Data Collection and Analysis

The subject wore IMU sensors (Myomotion, Noraxon Inc., USA) at the waist, thighs, calves and feet, which were used to measure the lower limb kinematics and trunk tilts during the trials. The IMU data was recorded at the PC at a rate of 100Hz and processed using the proprietary software (Myoresearch, Noraxon Inc.). Further processing of the data was carried out using MS Excel (Microsoft Inc., USA) to obtain symmetry ratio (SR) of the knee range of motion (ROM), and the root mean square (RMS) values of the medio-lateral (ML) and antero-posterior (AP) trunk tilts. The gait speed during all the trials was calculated based on the time taken by the subject to complete the measured 10m walk. The times were measured using a walk-through gate type system (Seed tech, Korea) that had one sensor placed at the start of the walking track and one at the end.

### 4. RESULTS

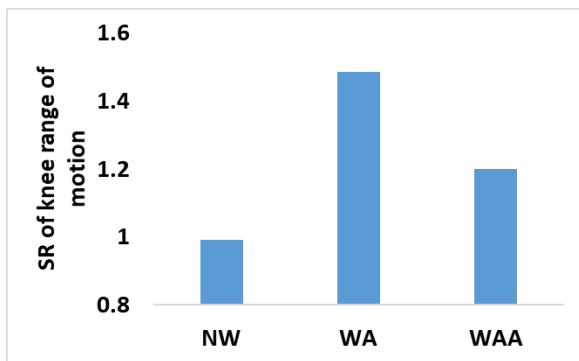
The subject did not report any discomfort in using the system. Table 3 presents the gait speed, knee ROM and RMS values of the ML and AP trunk tilts under all the trial conditions. Values presented here are the means of the values obtained from the two trials under each condition. The SR of knee ROM under different trial conditions are presented in Fig. 6.

These results show that addition of the weight at the ankle reduced the overall knee movement of the weighted leg. This resulted in increased knee ROM asymmetry between the legs. Addition of the ankle weight also decreased the gait speed while increasing the trunk tilts, especially in the AP direction. The increase in RMS of trunk tilts is indicative of reduced gait stability.



**Table 3.** Mean values of the outcome measures under the various test conditions.

| Measure              | Value  |        |        |
|----------------------|--------|--------|--------|
|                      | NW     | WA     | WAA    |
| Gait Speed (m/s)     | 0.940  | 0.623  | 0.751  |
| Left knee ROM (deg)  | 69.174 | 53.096 | 69.440 |
| Right knee ROM (deg) | 69.785 | 35.735 | 57.898 |
| RMS of ML tilt (deg) | 1.822  | 2.426  | 2.440  |
| RMS of AP tilt (deg) | 1.891  | 3.068  | 3.873  |



**Fig. 6.** SR of knee range of motion under the various test conditions.

The results also show that, as compared to the WA condition, use of the AKO in the WAA condition substantially improved the knee ROM. Furthermore, the SR of knee ROM results show that the AKO also helped to improve the symmetry between the legs. This is indicative of the AKO's capability to provide assistive torque to the knee joint in a manner that augments the torque generated by the leg muscles. The assistive torque provided by the AKO also helped to increase the subject's gait speed, which is considered as a favorable outcome during post-stroke gait rehabilitation. However, use of the AKO did not have an improving effect on the trunk tilts. In fact, the addition of the AKO's weight appeared to generate a small increase in the RMS of trunk tilts. This outcome warrants further improvement of the AKO design for future applications.

Overall, the results obtained through the pilot test indicate that the developed lightweight AKO is capable of providing sufficient amount of assistive torque in a pattern that augments the torque generated by the user's leg muscles. Therefore, this system holds promise as an assistive device for use during post-stroke gait rehabilitation. However, further studies with a large and diverse subject population are needed to gain more insights into the effects of using this system so that rehabilitation protocols that involve its utilization can be formulated.

## 5. CONCLUSION

In this paper, we presented the development of a lightweight AKO that is capable of providing assistive torque in a pattern that is similar to the natural pattern of the human leg. Furthermore, the developed AKO prototype was pilot tested with a young healthy subject walking under the disturbance of an added ankle weight. The results showed that use of the AKO resulted in substantial improvements in SR of knee ROM and gait speed. Therefore, the developed AKO holds potential for use in post-stroke gait rehabilitation, however further studies are required to better understand the implications of using this system in such an application.

## ACKNOWLEDGEMENT

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## REFERENCES

- [1] Pohl M, Werner C, Holzgraefe M, Krocze G, Wingendorf I, Hoölig G, Koch R, Hesse S. Repetitive locomotor training and physiotherapy improve walking and basic activities of daily living after stroke: a single-blind, randomized multicentre trial (DEutsche GAngrainerStudie, DEGAS). *Clinical rehabilitation*. 2007 Jan;21(1):17-27.
- [2] Lee, Hosu, et al. "A Novel Gait Assistance System Based on an Active Knee Orthosis and a Haptic Cane for Overground Walking." *International Conference on NeuroRehabilitation*. Springer, Cham, 2018.
- [3] Patel, S., et al.: Effects on normal gait of a new active knee orthosis for hemiparetic gait retraining. In: 28th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 1232–1235. IEEE, 2006.
- [4] Kalita B, Narayan J, Dwivedy SK. Development of active lower limb robotic-based orthosis and exoskeleton devices: a systematic review. *International Journal of Social Robotics*. 2020 Jun 23:1-9.
- [5] Li, Lifang, et al. "Improved walking ability with wearable robot-assisted training in patients suffering chronic stroke." *Bio-medical materials and engineering* 26.s1 (2015): S329-S340.
- [6] Pyo, Sanghun, Junwon Yoon, and Min-Kyun Oh. "A novel robotic knee device with stance control and its kinematic weight optimization for rehabilitation." *Robotica* 32.8 (2014): 1245-1263.
- [7] Bernhardt, Michael, et al. "Hybrid force-position control yields cooperative behaviour of the rehabilitation robot LOKOMAT." *Rehabilitation Robotics*, 2005. ICORR 2005. 9th International Conference on. IEEE, 2005.
- [8] Muratagic, Haris, Tyagi Ramakrishnan, and Kyle B. Reed. "Combined effects of leg length discrepancy and the addition of distal mass on gait asymmetry." *Gait & posture* 58 (2017): 487-492.